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Performance effects of functional knee brace removal and prolonged use in healthy male athlete: Lower extremity power, acceleration, speed, and agility

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ABSTRACT

Objectives: Knee braces were introduced to sports 30 years ago. However, knee brace use for non-contact anterior cruciate ligament injury prevention intervention remains contentious due to concerns about performance hindrances. Since knee brace use is a potential modifiable risk factor, we aimed to investigate the effect of discounting and continued functional knee brace (FKB) on lower extremity power-vertical jump (VJ), acceleration, speed, and agility performance.
Design: Prospective cohort crossover study.
Methods: Twenty-seven healthy male athletes performed seven tests, over six days of 12 test sessions (S), during three test conditions (non-braced, braced, and removed brace or continued brace use). This study focuses on VJ, acceleration, speed, and agility performance during S12 when athletes were randomly selected to remove the FKB after 17.5 h or continue using the FKB for 21.0 h.
Results: After brace removal, nonsignificant performance levels improved in the VJ (2.7 %; 95 % CI 52.5–62.8; Cohen's effect size (ES) = trivial), acceleration (1.8 %; 95 % CI 0.500–0.562; ES = small), and agility (0.5 %; 95 % CI 9.25–10.13; ES = trivial), while a nonsignificant slower speed was recorded (0.5 %; 95 % CI 1.81–1.95; ES = trivial). Continued brace use led to a nonsignificant performance improvement in all tests; VJ (3.1 %; 95 % CI 53.5–60.2; ES = small), acceleration (1.5 %; 95 % CI 0.511–0.561; ES = trivial), speed (1.0 %; 95 % CI 1.83–1.95; ES = trivial), and agility (1.8 %; 95 % CI 9.26–10.04; ES = trivial).
Conclusions: Removal of FKB led to improved performance in three performance tests, while continued brace use improved performance in all four tests.
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Practical implications

- Removing a functional knee brace (FKB) after 17.5 h of use led to an improvement in lower extremity power-vertical jump, acceleration-2 m, speed-10 m, and agility performance.
- Continuing use of an FKB for 21.0 h also led to an improvement in lower extremity power-vertical jump, acceleration-2 m, speed-10 m, and agility performance.
- However, a higher percentage and Cohen effect size measure performance improvements were observed with continued brace use for 21.0 h.

1. Introduction

Up to 96 % of primary non-contact anterior cruciate ligament (ACL) injuries occur by a non-contact mechanism (a single-leg landing resulting in increased loading of the knee joint at initial ground contact which, has been attributed to an upright stance and decreased knee joint flexion with increased quadriceps activation and simultaneous delayed hamstring activation),[1–3] and primary non-contact ACL injuries continue to be a concern for both sexes participating in team sports.[4,5] Furthermore, after primary non-contact ACL injury surgery, individuals

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face additional concerns regarding the reduced odds of returning to elite sport, increased risk of re-injury, injury to the contralateral knee joint,[6] and an increased risk of experiencing osteoarthritis.[7] These concerns increase as researchers have suggested that partial or complete ACL tears can also occur after repetitive near-maximal loading events (below the ultimate tensile strength of a healthy ligament), each causing enough microdamage to accumulate and combine to weaken the ligament.[3]

Based on the principles of injury prevention programs (IPPs), muscle preactivation/motor learning, and the identified mechanism of reducing primary non-contact ACL injury (lowering the load on the knee joint by increasing trunk/hip and knee flexion with simultaneous activation of the hamstring and deactivation of the quadriceps muscle groups), it would be assumed that there would have been a decrease in primary non-contact ACL injury rates.[3,8] However, published data suggest that the rate of primary non-contact ACL injury increased significantly in Australia,[9] in Norway,[10] and in the United States (US).[11] We suggest that this lack of reduction in the rate of primary non-contact ACL injury could be attributed to inherent principles of muscle physiology and neurology,[12] which do not allow alteration in the mechanism of primary non-contact ACL injury, suggesting that external knee joint support may be necessary to reduce the risk of primary non-contact ACL injury.

Two types of knee braces are recognized in sports, prophylactic (PKB) and functional (FKB).[13,14] PKBs are 'off-the-shelf' knee braces designed to prevent or reduce valgus and rotational stress and FKBs are custom-made braces designed to provide stability for ACL injured knees.[13–15] In 2023, the American Academy of Orthopaedic Surgeons Clinical Practice Guidelines provided a "Limited" recommendation due to "little or no evidence" on the potential benefits of PKB.[16] A recent retrospective analysis and a systematic review of knee brace prophylaxis suggested that the use of PKB may be associated with a reduced risk of "major" knee and primary non-contact ACL injury.[13,17] Furthermore, a recent publication reported that prolonged use of an FKB led to slower knee joint loading while performing a single-leg landing.[12] Despite the apparent preventative role, specifically in performing tasks utilizing the anaerobic alactic and lactic systems considered essential for team sports,[18] researchers have reported that a knee brace is not widely used for preventive intervention due to concerns about hindrance of athlete agility, sprint speed, and vertical jump performance.[19–21] In contrast, other researchers have suggested a negligible influence of knee brace use on performance in athletes with an uninjured knee joint.[20,22] However, most of these studies investigated the use of PKB, and a few studies evaluated the use and performance of FKB in lower extremity power-VJ (VJ),[20,22] speed,[19,20] and agility.[20] However, all studies evaluated the immediate effect of knee brace application on performance.[14] One study reported that while performing tasks requiring anaerobic alactic and lactic systems, after 14.0 h of FKB use, no statistically significant difference in performance level was recorded between non-braced and braced athletes.[23]

Lower extremity power-VJ, acceleration-2 m, speed-10 m, and agility performance are crucial in team sports.[18] Briefly, lower extremity power is considered vital for success, as it has been suggested to contribute significantly to developing basic game skills.[18] The importance of high acceleration is highlighted because the first three steps of acceleration can play a vital role in achieving maximum sprint velocity in field sports.[24] Maximal speed capabilities are required to succeed in attacking and defensive plays.[18] Agility skills (require multidisciplinary attributes; defined in this study to rapidly change body direction and position in the horizontal plane without loss of speed[18,25]) are suggested to be an essential prerequisite for elite players and influence team game results.[18]

Therefore, an understanding of the removal of an FKB after use and the prolonged use of an FKB by healthy athletes on VJ, acceleration, speed, and agility performance is warranted, which could assist practitioners in addressing the concerns presented in the current literature. Accordingly, this study aimed to investigate whether the levels of VJ, acceleration, speed, and agility performance of healthy athletes were hindered in removing an FKB after 17.5 h of use and with continued use of an FKB for 21 h.

2. Methods

This study was approved by a University Ethics Committee (06/007) and a University Clinical Research Ethics Board (C06/0480) and was conducted in accordance with the ethical standards of the Declaration of Helsinki. Following the respective ethical procedures, all athletes provided written informed consent to participate in this study.

2.1. Study population

Twenty-seven male athletes, provincial basketball (n = 14) and national field hockey (n = 13), with no history of knee injury or no previous history of knee brace use, volunteered for this study. Although some may suggest that female athletes are at greater risk for a primary non-contact ACL injury, published data from Sweden,[26] the US,[11] and specific to college athletes (National Collegiate Athletic Association)[27] reported that men were more likely to sustain a primary non-contact ACL injury. Therefore, to ensure adequate study priori power for the performance tests and test conditions used in this study, data were collected only from healthy male athletes. Some may also suggest that basketball is a jumping/landing sport; however, time motion analysis data have established that jumping/landing activity represented only 2.1 ± 0.3 % of all required movements during "live time" gameplay and decreased at lower levels of competition.[28] The mean age (years), weight (kg), and height (cm) of the athletes were 19.4 ± 3.0 (range 17–26), 80.0 ± 9.1 (range 63.6–103.6), and 182.6 ± 6.8, respectively. Detailed characteristics of the athletes for each test condition are presented in Table 1. The history of knee injuries and brace use was verified by the respective athletic director. Each athlete used a custom-fitted FKB on their dominant leg; which was defined as the stance, or push-off leg, during a running jump. The selection of FKB, Extreme® (Össur, Canada), was based on published data.[29] The characteristics of FKB for each test condition are detailed in Table 1. The FKB used consisted of a hinge, post, and shell design that, according to the brace manufacturer, provided anterolateral and posteromedial stability, and the derotational strap maintained anteromedial rotatory stability. To ensure that there was no conflict of interest, the FKBs were purchased.

Table 1
Athlete and functional knee brace characteristics for each test condition.

	n	Athletes			FKB
		Age (years)	Weight (kg)	Height (cm)	Weight (grams)
		Mean ± SD			
Test conditions 1 and 2, S1–5 and S7–11, respectively	27	19.4 ± 3.0	80.0 ± 9.1	182.6 ± 6.8	759 ± 97.1[a]
Test condition 3, S12-*removed brace*	12	18.9 ± 3.4	82.2 ± 11.0	184.8 ± 6.1	NA
Test condition 3, S12-*continued brace use*	15	19.8 ± 2.7	78.2 ± 7.2	180.9 ± 7.0	763.3 ± 91.5

cm: centimeters; FKB: functional knee brace; kg: kilograms; NA: not applicable; SD: standard deviation; S: test session.
[a] The FKB weight is only applicable to test sessions 7–11.

2.2. Study design

This prospective closed cohort repeated-measure crossover study followed previously published methodology, detailed in the original study,[23] and was part of a larger experiment consisting of athletes performing seven tests, during three test conditions (non-braced, braced, and removed brace, or continued brace use), for six days of 12 test sessions (S) and a total of 38.5 h (Fig. 1). Acceleration, agility, lower extremity power-vertical jump, and speed performance data, days 1–3, S1–5, and braced test condition, days 4–6, S7–11 were previously published.[23] Briefly, performance levels decreased in all performance tests when an FKB was used initially. However, after 14.0 h of FKB use, no statistically significant differences in performance levels were recorded between non-injured non-braced and braced athletes.

The repeated-measure crossover design allowed the control and experimental groups to be matched for ability and physiological performance levels (preventing possible performance bias) for all tests completed during each testing session (Fig. 1). The repeated-measure design also ensured adequate study power (see Statistical analysis) and allowed continued accommodation/familiarization with the use of FKB, the latter being a key part of the study. This paper concentrates on the levels of VJ, acceleration, speed, and agility performance during S12 non-braced (removed brace after 17.5 h) and braced (continued brace use for 21.0 h) conditions. We selected these periods because previous work had suggested that the use of FKB for 17.5 h did not significantly hinder performance in these high-intensity tests.[23]

Tests performed each test session (S)

(1) Drop jump kinetics-time to peak vertical ground reaction force (PVGRF) and PVGRF,

(2) Lower body power-vertical jump,

(3) Acceleration (2-meters),

(4) Sprint (10-meters),

(5) Agility-T-test,

(6) Anerobic endurance-Repeated High Intensity Shuttle Test, and

(7) **Aerobic capacity-Léger Beep Test**

	Testing duration ***(hours)***	***Braced duration*** ***(hours)***
Test condition 1: Non-braced (n=27)		
Day 1, S1-2	7.0	
Day 2, S3-4	14.0	
Day 3, S5	*17.5*	
Day 3, S6 - REST		
Test condition 2: Braced (n=27)		
Day 4, S7-8		7.0
Day 5, S9-10		14.0
Day 6, S11		*17.5*
Test condition 3:		
Removed brace (n=12); Day 6, S12	***21.0***	
Continued brace use (n=15); Day 6, S12		***21.0***

Fig. 1. Test schedule and duration. Athletes performed seven tests during each test session (S). Each test session lasted for 3.5 h for a total of 38.5 h of testing over six days. After completing 17.5 h of non-braced testing (test condition 1, days 1–3, S1–5), the athletes completed 17.5 h of braced testing (test condition 2, days 4–6, S7–11), and then the athletes were randomly selected for the groups for test condition 3, day 6, S12, 'Removed brace' or 'Continued brace use'. This paper concentrates on the levels of lower extremity power-VJ, acceleration, speed, and agility performance during the conditions non-braced (S12, removed brace after 17.5 h) and braced (S12, continued brace use for 21.0 h) (highlighted and in *italic* font). Statistical analysis was completed between test condition 1, non-braced, day 3, S5 (non-braced testing for 17.5 h, in *italic* font) and test condition 3, day 6, S12 (removed brace after 17.5 h) as well as between test condition 2, day 6, S11 (braced for 17.5 h, in *italic* font) and test condition 3, day 6, S12 (continued brace use for 21 h).

Table 2
Priori power analysis.

Performance test	Difference[a]	SD	n	Power	n	Power	n	Power
Lower extremity power-VJ (cm)	2.67	4.46	10	0.394	20	0.719	25	0.819
Acceleration-2 m (sec)[b]	0.01	0.01	10	0.616	20	0.892	25	0.940
Speed-10 m (sec)	0.09	0.09	10	0.803	20	0.989	25	0.998
Agility-T-Test (sec)	0.11	0.22	10	0.293	20	0.565	25	0.670

cm: centimeters; m: meters; SD: standard deviation; sec: seconds; VJ: vertical jump.
[a] Difference between non-braced and braced performance measures.[22]
[b] Data were not collected for the acceleration-2-m test in the pilot study. Difference between non-braced and braced performance measures calculated using original study data.[22] The significance level was set to 95 %, and the statistical power was set to 0.80.

2.3. Study procedure and data collection

Detailed testing and data collection procedures for each performance test are presented in the original publication.[23] Briefly, after a 20-minute warmup, the athletes completed the seven performance tests listed in Fig. 1, and three trials were performed for the VJ, acceleration (2 m), speed (10 m), and agility tests. Although the relationship between lower extremity power and VJ appears clear,[30] there is a belief that the evaluation of VJ and power is not identical.[31] However, power depends on maximum strength and an increase in the latter variable is related to an improvement in power ability, the shared relationship between power production during a specific movement and a maximum repetition of this movement is approximately 50 % or more, and tends to increase as the weight lifted increases.[30] Specifically regarding agility, although some may suggest that agility should be defined as "a rapid whole-body movement with a change of velocity or direction in response to stimulus",[32,33] required in 'on-field' game situations, the objective of this study was to evaluate performance when removing an FKB and continued use of an FKB, thus no "response to stimulus" was required. Data were collected using Brower ID XS electronic gates (Brower Timing Systems, Draper, Utah, US) set at a height of 75 cm. The athletes received a 1-minute rest between each trial.

2.4. Statistical analysis

Following previously published data,[23] a priori power analysis was performed for each performance test (Table 2). Unpaired *t*-tests were performed for the group means of VJ, acceleration (2 m), speed (10 m), and agility test data between non-braced S5 (test condition 1, day 3, non-braced testing for 17.5 h, n = 27) and S12 (test condition 3, day 6, removed brace after 17.5 h, n = 12) as well as between S11 (test condition 2, FKB use (braced) for 17.5 h, n = 27) and S12 (test condition 3, day 6, continued brace use for 21 h, n = 15). Furthermore, the respective 95 % confidence intervals (95CIs), the percentage difference, and Cohen's effect size (ES) with the corresponding descriptors, trivial = 0–0.19, small = 0.20–0.49, medium = 0.50–0.79, and large > 0.80[34] were calculated. The significance of all tests was determined at the level of $p \leq 0.05$. Data were analyzed using SPSS for Windows, version 19.0 (SPSS Inc., Chicago, IL, US).

3. Results

Statistical significance was not achieved in the four performance tests. Upon removal of the FKB after 17.5 h of use (test condition 3,

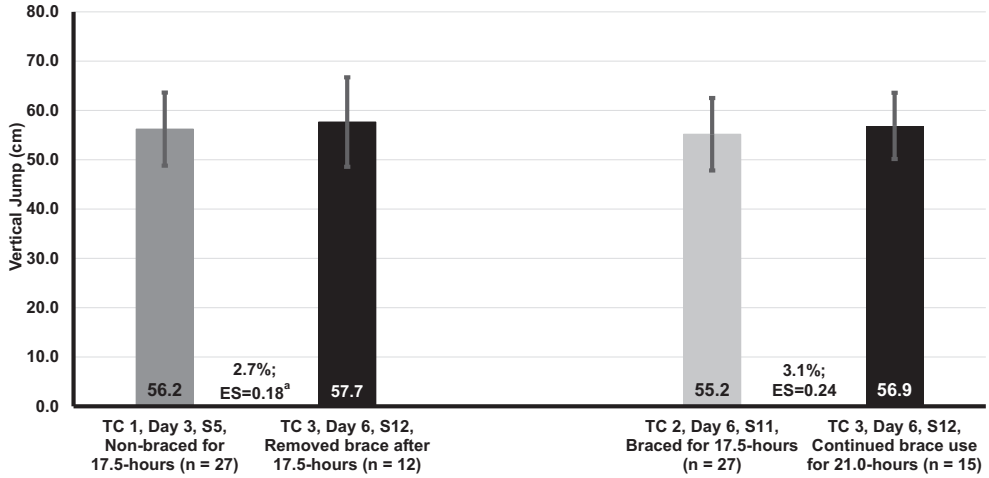

Fig. 2. Performance of lower body power-vertical jump.[b] Comparison between test condition (TC) 1, day 3, session (S) 5, non-braced for 17.5 h (95 % confidence interval (CI) = 53.4–59.1) and TC 3, day 6, S12, removed functional knee brace (FKB) after 17.5 h (CI = 52.5–62.8) resulted in a non-significant higher vertical jump (p = 0.704). Evaluation between TC 2, day 6, S11, after using FKB for 17.5 h (CI = 52.4–57.9) and TC 3, day 6, S12, continued use of FKB for 21 h (CI = 53.5–60.2) led to a non-significant higher vertical jump (p = 0.661). [a]Cohen effect size with the corresponding descriptors, trivial: 0.0–0.19, small: 0.20–0.49, medium: 0.50–0.79, and large: >0.80.[34] [b]Test performed using the Vertec vertical jump apparatus (Sports Imports, Hilliard, Ohio, US), and arms were allowed to swing freely.

day 6, S12, removed brace) compared to test condition 1, day 3, S5, non-braced for 17.5 h, there was a percent performance improvement (−0.5 % to 2.7 %) in all four performance tests (Figs. 2–4). The continued use of FKB for 21 h (test condition 3, day 6, S12) compared to test condition 2, day 6, S11, braced (use of FKB for 17.5 h), also led to a percent improvement in performance in the four tests (Figs. 2–4). However, a higher percentage of performance improvement (1.0 % to 3.1 %) was observed with continued FKB use for 21 h (test condition 3, day 6, S12) compared to FKB use for 17.5 h (test condition 2, day 6, S11) in the VJ, speed-10 m, and agility tests (Figs. 2–4). The opposite was true in the acceleration-2 m test (Fig. 3). This tendency was also observed with Cohen's ES measure (ranged between trivial and small); greater performance improvement with continued FKB use for 21 h (condition 3, day 6, S12) versus removing the FKB after 17.5 h of use (test condition 3, day 3, S12, removed brace) (Figs. 2–4).

4. Discussion

The most important findings of the present study were that, in healthy male athletes, in the four performance tests (VJ, acceleration-2 m, speed-10 m, and agility), regardless of removing the FKB after 17.5 h of use or continued use of the FKB for 21.0 h, led to improvement in performance. However, a higher percentage and Cohen's ES measure performance improvements were observed with continued brace use for 21.0 h.

Since the implementation of IPPs,[3,8] it would be assumed that there would have been a decrease in ACL injury rates. However, published data suggests that the rate of primary non-contact ACL injury has not decreased,[3,8] as data, between 1999 and 2021, from Australia, Norway, and the US reported that the rate of primary non-contact ACL injury and surgery increased between 22 % and 50 %[9–11] and the elevated

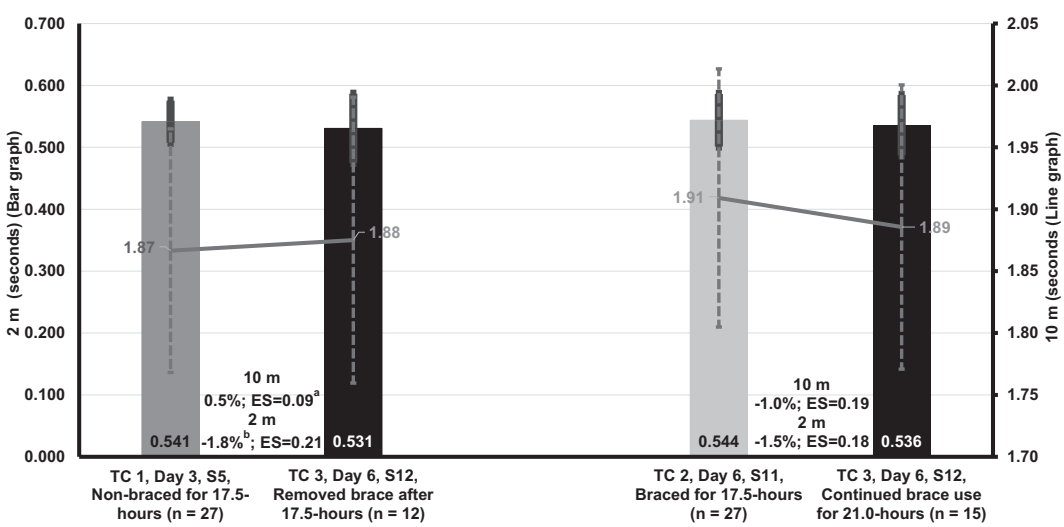

Fig. 3. Performance of acceleration (2 m) and sprint (10 m). Acceleration – the evaluation between test condition (TC) 1, day 3, session (S) 5, non-braced for 17.5 h (95 % confidence interval (CI) = 0.525–0.557) and TC 3, day 6, S12, removed functional knee brace (FKB) after 17.5 h (CI = 0.500–0.562) resulted in non-significant faster performance (p = 0.822). Comparison between TC 2, day 6, S11, after using the FKB for 17.5 h (CI = 0.528–0.560) and TC 3, day 6, S12, continued FKB use for 21 h (CI = 0.511–0.561) led to a non-significant faster acceleration (p = 0.763). Sprint – comparison between test condition (TC) 1, day 3, S5, non-braced for 17.5 h (CI = 1.83–1.91) and TC 3, day 6, S12, removed FKB after 17.5 h (CI = 1.81–1.95) resulted in non-significant slower performance (p = 0.384). Evaluation between TC 2, day 6, S11, after using the FKB for 17.5 h (CI = 1.87–1.95) and TC 3, day 6, S12, continued FKB use for 21 h (CI = 1.83–1.95) led to a non-significant faster sprint performance (p = 0.653). [a]Cohen effect size with the corresponding descriptors, trivial: 0.0–0.19, small: 0.20–0.49, medium: 0.50–0.79, and large: >0.80.[34] [b]The negative performance percentage indicates a performance improvement.

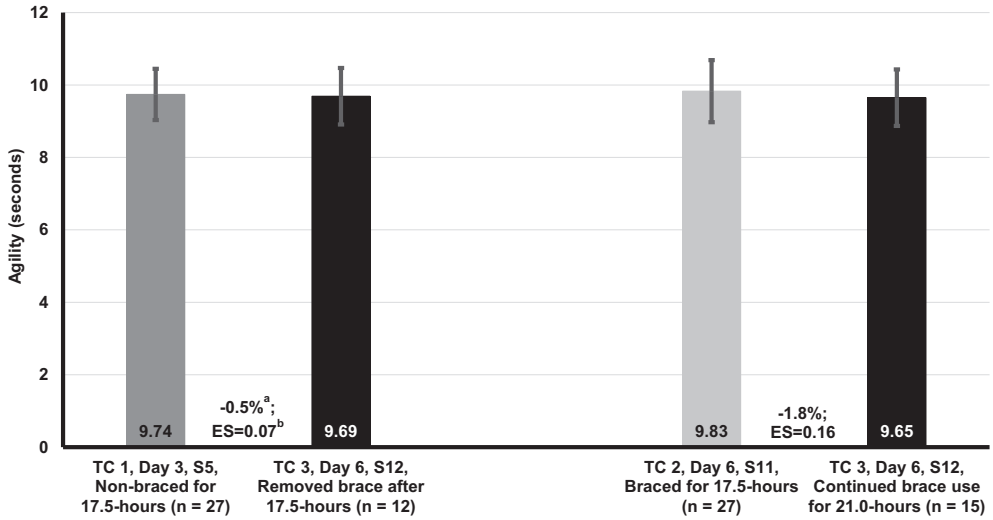

Fig. 4. Performance of agility (T-test). Comparison between test condition (TC) 1, day 3, session (S) 5, non-braced for 17.5 h (95 % confidence interval (CI) = 9.47–10.01) and TC 3, day 6, S12, removed functional knee brace (FKB) after 17.5 h (CI = 9.25–10.13) resulted in non-significant faster performance (p = 0.467). Evaluation between TC 2, day 6, S11, after using the FKB for 17.5 h (CI = 1.87–1.95) and TC 3, day 6, S12, continued use of FKB for 21 h (CI = 9.26–10.04) led to non-significant faster performance in the agility test (p = 0.778). [a]The negative performance percentage indicates a performance improvement. [b]Cohen effect size with the corresponding descriptors, trivial: 0.0–0.19, small: 0.20–0.49, medium: 0.50–0.79, and large: >0.80.[34]

risk of primary non-contact ACL injury continues to be of concern for both sexes.[4,5] We suggest that the reported higher rate of primary non-contact ACL injury could be attributed to the inherent principles of muscle physiology and neurology detailed in previous work.[12]

A retrospective analysis and systematic analysis suggested that the use of PKB may be associated with a reduced risk of "major" knee and primary non-contact ACL injury[13,17] and a recent repeated-measures cross-over study revealed that prolonged use of an FKB led to slower loading of the knee joint while performing a single-leg landing.[12] However, a knee brace is not widely used for preventive intervention due to concerns about impeded performance.[19–21]

Our study findings of no significant differences in VJ, acceleration-2 m, speed-10 m, and agility tests in the removal of the FKB after 17.5 h of use address the previously published "theory that bracing weakens the knee".[35,36]

The results of the present study of no significant differences in the four performance tests with continued use of the FKB brace are inconsistent with previously published results of slower performance in the 40-yard (36.6-m) speed[19,20] and agility tests[20] or limited impedance in performance in VJ.[20,22] However, most of these studies investigated the use of PKB and all studies evaluated the immediate effect of the application of a knee brace on performance.[14] It should be noted that although Veldhuizen et al. evaluated performance after using a knee brace for 28 days, the authors did not specify the duration of use of the knee brace,[36] therefore, the results are not included in our paper.

Published work has suggested that knee brace straps cause compression of the lower extremity soft tissue, which could lead to increased energy expenditure and premature muscle fatigue due to tissue hypoxia.[21] However, other researchers have reported, using animal studies, that external compression more easily compromises blood flow circulation in the skin than muscle circulation,[37] and, in human participants, high intramuscular pressure values were not always associated with hypoxia in muscle tissue due to external soft tissue compression.[38] Although we did not measure the effects of muscle fatigue, if 38.5 h of testing on six consecutive days (Fig. 1) were considered to result in muscle fatigue, it could be hypothesized that the higher percentage and Cohen's ES measure performance improvements with continued brace use could help clarify concerns about premature fatigue presented by Styf.[21] However, the four performance tests used in this study focused on tasks utilizing the anaerobic alactic and lactic systems, and therefore research is needed, with prolonged use of an FKB, that challenges the anaerobic endurance and aerobic capacity systems.

The results of this study of improved performance after 21.0 h of brace use (Figs. 2–4) could be attributed to greater proprioceptive feedback, as data from braced patients with ACL-deficiency and post-ACL surgery braced patients have reported improved levels of performance, although these knee joints had proprioceptive deficits (tactile afferents) that inhibited the input to the central nervous system (CNS).[39] The importance of proprioception and tactile afferents was exemplified in patients with Hereditary Sensory and Autonomic Neuropathy (HSAN) Type III, who had no evidence of cerebellar atrophy, normal muscle tone, and strength, but their tendon- or M- and H-reflexes were absent, which resulted in significant gait ataxia.[40] On application of tape bilaterally to the HSAN III knee joints, a significant improvement in proprioceptive gait performance was observed, but not in controls.[40] If one were to broaden the CNS proprioceptive dogma and apply it to a non-injured knee joint, with normal M- and H-reflex proprioception, then perhaps after 21 h of FKB use, healthy braced athletes experienced increased proprioceptive feedback, resulting in a higher percentage and Cohen ES measure performance improvement. Our broadened CNS proprioceptive dogma is supported by published data from functional magnetic resonance imaging that reported significantly higher activation in the primary sensorimotor cortex (paracentral lobule of the frontal lobe) and the superior parietal lobule (Brodmann area 5) when knee flexion-extension movement was performed from 0 to 90° with the use of a "tight" knee brace compared to the use of an elastic knee sleeve and without the use of a knee brace or sleeve.[41]

This study may be limited by the performance ability of each athlete. Although each athlete was a member of a provincial or national team, the athletes differed in their individual intensity and motivation levels. The competitive nature of these athletes may have caused peer competition that led to increased motivation and possibly influenced increased performance. Since there is still a debate regarding sex[4,5,13] and the risk and rate of primary non-contact ACL injury,[9–11] to ensure adequate study priori power for most performance tests and conditions, data were collected only in healthy male athletes. These data may not apply to female athletes and/or healthy athletes who may be at risk, due to inherent anatomical and/or biomechanical factors, of sustaining a primary non-contact ACL injury. A larger cohort and/or a longer duration of FKB use may lead to greater adaptation/familiarization of FKB, which may further improve the respective performance measure(s). To eliminate testing conflicts of interest or product testing bias, no external funding was pursued or used to purchase the FKBs, and the authors do not endorse any FKBs.

5. Conclusions

This study demonstrated that performance improvement levels were similar after removing the FKB after 17.5 h of use or continued use of the FKB for 21.0 h when performing the VJ, acceleration, speed, and agility tests. However, a higher percentage and Cohen's ES measure performance improvements were noted in the VJ, speed, and agility tests with continued brace use for 21.0 h. The results of this study should be of assistance to healthcare professionals in discussing possible concerns and potential benefits about the use of an FKB and performance concerns with parents and healthy athletes.

CRediT authorship contribution statement

N.R. conceptualized and conducted the study. N.R., J.E.T., R.L.S., and B.N. contributed to the study design and statistical analysis. All authors made substantial contributions to writing, reviewing, and editing, and provided final approval of the manuscript.

Confirmation of ethical compliance

This study was approved by the University of Otago, New Zealand Ethics Committee (reference code: 06/007) and the University of British Columbia Clinical Research Ethics Board, Canada (Certificate Approval Number: C06-0480), and athletes gave their written informed consent to participate in the study before participating.

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Declaration of interest statement

The authors declare that they have no conflicts of interest with respect to the research, authorship, and publication of this article.

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